

What are the causes and risk factors?

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- **Type 1 diabetes is caused by an absolute insulin deficiency, usually resulting from autoimmune destruction of the insulin-producing beta cells in the pancreas.**
- **Risk factors include:**
 - **Genetic factors** — type 1 diabetes is a heritable polygenic disease.
 - A sibling of a person with type 1 diabetes has 6–7% risk of developing it themselves. This risk increases to 30–70% in identical twins.
 - The risk for a child is 1–9% if a parent has type 1 diabetes.
 - Several different alleles (alternative forms of a gene) and haplotypes (combinations of alleles) have been associated with either an increased or decreased risk of type 1 diabetes. However, the mechanisms by which these haplotypes interact and alter risk are not completely understood.
 - **Environmental factors** — there is evidence that in genetically susceptible people, certain environmental factors can trigger the development of autoimmunity to the pancreatic beta cells.
 - Environmental factors associated with type 1 diabetes include diet, vitamin D exposure, obesity, early-life exposure to viruses associated with islet inflammation (such as enteroviruses), and decreased gut-microbiome diversity.
 - The importance of environmental risk factors (regardless of genetic background risk) are highlighted by the large differences in the incidence of type 1 diabetes in genetically similar populations that are separated by socioeconomic borders and by the increasing incidence of type 1 diabetes in genetically low-risk people.

[[NICE, 2015a \(/topics/diabetes-type-1/references/\)](#); [DiMeglio 2018 \(/topics/diabetes-type-1/references/\)](#); [WHO, 2019 \(/topics/diabetes-type-1/references/\)](#); [Norris, 2020 \(/topics/diabetes-type-1/references/\)](#)]

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